SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Dry Run Evaluation

Course Code:	CSA17	**Course Title:**	Artificial Intelligence
CO Assessed:	CO1 – Apply (BL3)	**Assessment:**	Assessment Tool 2 – Dry Run Evaluation
Weightage:	35% of CIA	**Duration / Marks:**	60 mins / 100 marks
CO1 Statement:	Apply AIbasics such as intelligent agents and uninformed search methods to solve problemseffectively. (BL3)		
Student Name:	S.Yamini	**Reg. No.:**	1973260015
Section / Batch:	2026- 3rd	**Date:**	08.10.2026

Code of Conduct

I S.Yamini, 1973260015 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will resultin disciplinaryaction.*

Signature of the Student: S.Yamini

Instructions

- Readeachquestion carefully; answer all four questions (25 marks each, 100 in total).
- Showeverydry-run step: the initial queue, stack, priority queue or belief state, each iteration, your assumptionsand a clear final answer, and finish with the interpretation asked for.
- Followtheconventions stated in each question (alphabetical expansion, explored set, goal test on removalfromthe frontier, tie-breaking) and present tables neatly.
- Marksfollowthe Dry Run Evaluation rubric: understanding of the search problem, state expansion and dryrunexecution, application of the search algorithm, accuracy of the solution path, and presentationand logical reasoning.
- UploadinGitHub and GCR (question paper with Reg. No., Name, answers, rubrics and evaluation pageattheend). Duration: 1 hour (15 minutes per question).

Section / Topic	Focus	Question Nos.
1. Foundations and History of AI	Four approaches toAI, Turing Test, eras and milestones	**Q2**
2. Intelligent Agents: Agents and Environments	Agent, percepts, actions, agent function and program, PEAS	**Q1**
3. The Concept of Rationality	Performance measure, percept sequence, rational vs omniscient	**Q1**
4. The Nature of Environments	Task-environment properties and their classification	**Q1**
5. The Structure of Agents	Simple reflex and model-based reflex agent programs	**Q1**
6. Problem-Solving Agents	Goal formulation, problem formulation, search, execution	**Q2**
7. Problem Formulation	States, initial state, actions, goal test, path cost; belief states	**Q2, Q4**
8. Searching for Solutions	Search graph, frontier, explored set, node expansion	**Q3**

9. Uninformed Search Strategies	Dry runs of BFS, DFS and uniform-cost search	**Q3**
10. Search with Partial Information	Sensorless problems, belief states, percept update	**Q4**

Annexure A – Dry Run Evaluation

Question 1: Agents, Environments and Rationality – Dry Run of Agent Programs

Topic:Intelligentagents,rationality,natureofenvironments,structureofagents | 15minutes

A vacuum-cleaner agent lives in a world withtwosquares, Aand B. At eachtimestepitperceives [its location, status of its own square] and performs one action: Suck, Right, LeftorNoOp.Theagent starts in square A, both squares are initially Dirty, and dirt never reappears.
Performance measure over T = 6 time steps: +10 for every square cleaned(aSuckonaDirty square) and −1 for every Left or Right move.
Agent X (simple reflex): if the current square is Dirty, Suck; otherwise movetotheothersquare.
Agent Y (model-based reflex): keeps an internal state of the squares it knowstobeclean.Ifthe current square is Dirty, Suck; otherwise, if both squares are known clean, NoOp; otherwisemovetothe other square.

(a) Write the PEAS description (Performance measure, Environment, Actuators, Sensors) of this agent. **[4 Marks]**

(b) Classify the task environment on observability, number of agents, determinism, episodic/sequential, static/dynamic and discrete/continuous, with a one-phrase justification each. **[4 Marks]**

(c) Dry-run both agents for all 6 time steps. Complete the table (percept, action, and Y's internal state after the step). **[9 Marks]**

Step	X: Percept	X: Action	Y: Percept	Y: Action	Y: Known-Clean Squares
1					
2					
3					
4					
5					
6					

(d) Compute the total performance of X and of Y over the 6 steps. **[3 Marks]**

(e) Interpret: which agent is rational? Both agents receive the identical percept [B, Clean] at step 4 – explain why they act differently, using the structure of the two agent programs, and distinguish a rational agent from an omniscient agent. **[5 Marks]**

Answers – Question 1

(a) PEAS

Performance: +10 for each square cleaned; −1 for each Left/Right move over 6 steps.
Environment: two-square world (A, B), with dirt initially in both squares and no dirt reappearing.
Actuators: Suck, Left, Right, NoOp. Sensors: current location and dirt status of the current square.

(b) Task environment classification

- Observability: Partially observable — the agent senses only its current square.
- Agents: Single-agent — only the vacuum agent acts.
- Determinism: Deterministic — each action has a predictable result.
- Episodic/sequential: Sequential — current actions affect later states.
- Static/dynamic: Static — dirt does not reappear and the world changes only through actions.
- Discrete/continuous: Discrete — locations, dirt states, actions and time steps are discrete.

(c) Dry run

Step	X Percept	X Action	Y Percept	Y Action	Y Known-Clean
1	[A, Dirty]	Suck	[A, Dirty]	Suck	{A}
2	[A, Clean]	Right	[A, Clean]	Right	{A}
3	[B, Dirty]	Suck	[B, Dirty]	Suck	{A,B}
4	[B, Clean]	Left	[B, Clean]	NoOp	{A,B}
5	[A, Clean]	Right	[A, Clean]	NoOp	{A,B}
6	[B, Clean]	Left	[B, Clean]	NoOp	{A,B}

(d) Total performance

X: 2 cleanings × 10 − 4 moves × 1 = 16.
Y: 2 cleanings × 10 − 1 move × 1 = 19.

(e) Interpretation

Yismore rational here because it uses memory of previously cleaned squares and avoids unnecessary movement. At step 4 both receive [B, Clean], but X is a simple reflex agent and moves to the other square whenever the current square is clean, whereas Y knows both squares are already clean and chooses NoOp. A rational agent chooses the action expected to maximize performance from the information available; an omniscient agent would know the actual future/state outcomes perfectly.

Question 2: Foundations and History of AI; Problem-Solving Agent and Problem Formulation

Topic:FoundationsandhistoryofAI,problem-solvingagents,problemformulation | 15minutes

PartA.Complete thetable byclassifyingeachsystem or idea underone ofthe fourapproachestoAI (thinking humanly, acting humanly, thinking rationally, acting rationally) and naming the era or milestone of AI history to which it belongs.

(a) Complete the table. **[8 Marks]**

System / Idea	Approach to AI	Era / Milestone
Turing Test proposed (1950)		
General Problem Solver (Newell and Simon, 1957)		
Aristotle's syllogisms and formal logic		
An agent that does the right thing to maximise expected performance		

Part B. A problem-solving agent has two jugs: X of capacity 4 litres and Y of capacity 3 litres, both initially empty and without measuring marks. It may fill a jug from a tap, empty a jug on the ground, or pour one jug into the other until the source is empty or the target is full. The goal is to have exactly 2 litres in jug X.

(b) Formulate the problem: state representation, initial state, actions, transition model, goal test and path cost. **[6 Marks]**

(c) Generate all successor states of (0, 0) and of (4, 0) (leave out actions that leave the state unchanged). **[3 Marks]**

(d) How many states does the state space contain? Write a shortest solution as a sequence of states and give its path cost. **[3 Marks]**

(e) Interpret: describe the four phases (goal formulation, problem formulation, search, execution) of a problem-solving agent for this task, and explain why the agent can execute the found solution without looking at new percepts. **[5 Marks]**

Answers – Question 2

(a) Approaches to AI and milestones

• Turing Test proposed (1950):Acting humanly — milestone: Turing Test / early AI.
• General Problem Solver (Newell and Simon, 1957): Thinking humanly — milestone: early symbolic AI / cognitive simulation.
• Aristotle's syllogisms and formal logic: Thinking rationally — milestone: foundations of formal logic.
• An agent that does the right thing to maximise expected performance: Acting rationally — milestone: rational-agent approach to AI.

(b) Problem formulation

• State representation:(x,y), where x is the amount in the 4-L jug X and y is the amount in the 3-L jug Y.
• Initial state: (0, 0).
• Actions: Fill X, fill Y, empty X, empty Y, pour X→Y, and pour Y→X.
• Transition model: Apply the selected action until a jug is full/empty as appropriate.
• Goal test: x = 2 litres.
• Path cost: 1 per action; total path cost is the number of actions.

(c) Successors

From (0,0): (4,0) by Fill X; (0,3) by Fill Y.
From (4,0): (0,0) by Empty X; (1,3) by Pour X→Y.

(d) State-space size and shortest solution

Statespace:5× 4 =20 possible states, because X can contain 0–4 L and Y can contain 0–3 L.
Shortest solution: (0,0) → (0,3) → (3,0) → (3,3) → (4,2) → (0,2) → (2,0). Path cost = 6.

(e) Four phases

Goalformulation: decide that X must contain exactly 2 L. Problem formulation: define states, actions, transitions, goal test and cost. Search: find a least-cost sequence of jug operations. Execution: perform the selected actions in order. The agent can execute the solution without new percepts because the jug actions and transitions are deterministic and the complete initial state is known.

Question 3: Searching for Solutions – Uninformed Search Strategies

Topic: Searching for solutions, BFS, DFS, uniform-cost search | 15 minutes

Consider the weighted undirected graph below. The start node is S and the goal node is G. Use graph search with an explored set; expand neighbours alphabetically, test the goal when a node leaves the frontier, and break ties alphabetically. BFS and DFS ignore edge costs while searching, but path costs are measured with them.

Edge	Cost	Edge	Cost
S – A	1	B – D	1
S – B	4	C – G	6
A – C	2	D – G	2
A – D	5		

(a) Perform BFS from S (queue initially [S]). Do not add a node that is already explored or already in the queue. **[6 Marks]**

Iteration	Node Dequeued	Queue after Expansion	Explored Set
1			
2			
3			
4			
5			
6			

(b) Perform DFS from S using a stack (initially [S]); push children so that the alphabetically first child is popped first. Write the stack with its top on the left. **[5 Marks]**

Iteration	Node Popped	Stack after Pushing Children
1		
2		
3		
4		

(c) Perform uniform-cost search from S. Write queue entries as Node:g(parent) and replace an entry whenever a cheaper path to a frontier node is found. **[7 Marks]**

Iteration	Node Expanded	g(n)	Priority Queue after Expansion
1			
2			
3			
4			
5			
6			

(d) Complete the comparison table. **[4 Marks]**

Strategy	Path Found	Path Cost	Nodes Removed from Frontier	Optimal Here?
BFS				
DFS				

Strategy	Path Found	Path Cost	Nodes Removed from Frontier	Optimal Here?
UCS				

(e) Interpret: explain why UCS removed as many nodes as BFS yet found a cheaper path, why DFS needed the fewest nodes here but cannot be trusted in general, and why an explored set is needed in graph search (use iteration 3 of BFS as an example). **[3 Marks]**

Answers – Question 3

(a) BFS

1. Dequeue S → Queue [A,B] Explored {S}
2. Dequeue A → Queue [B,C,D] → Explored {S,A}
3. Dequeue B → Queue [C,D] → Explored {S,A,B}
4. Dequeue C → Queue [D,G] → Explored {S,A,B,C}
5. Dequeue D → Queue [G] → Explored {S,A,B,C,D}
6. Dequeue G → Goal reached.

Path: S → A → C → G. Path cost = 1 + 2 + 6 = 9.

(b) DFS

1. Pop S → Stack [A,B]
2. Pop A → Stack [C,D,B]
3. Pop C → Stack [G,D,B]
4. Pop G → Goal reached.

Path: S → A → C → G. Path cost = 9.

(c) Uniform-cost search

1. Expand S, g=0 → PQ [A:1(S), B:4(S)]
2. Expand A, g=1 → PQ [C:3(A), B:4(S), D:6(A)]
3. Expand C, g=3 → PQ [B:4(S), D:6(A), G:9(C)]
4. Expand B, g=4 → PQ [D:5(B), G:9(C)]
5. Expand D, g=5 → PQ [G:7(D)]
6. Remove G, g=7 → Goal reached.

Path: S → B → D → G. Path cost = 4 + 1 + 2 = 7.

(d) Comparison

• BFS: Path S-A-C-G; cost 9; 6 nodes removed; Optimal here: No.
• DFS: Path S-A-C-G; cost 9; 4 nodes removed; Optimal here: No.
• UCS: Path S-B-D-G; cost 7; 6 nodes removed; Optimal here: Yes.

(e) Interpretation

UCS removed as many nodes as BFS in this graph, but it considered edge costs and therefore found the cheaper path of cost 7. DFS reached the goal after only 4 removals because it followed the A→C branch first, but DFS is not guaranteed to find an optimal solution. An explored set prevents repeated expansion and cycles; for example, at BFS iteration 3, D is already in the queue, so B must not add D again.

Question 4: Search with Partial Information – Belief States

Topic: Problemformulation,searchwithpartialinformation(sensorlessandpercept-based belief-state search) | 15 minutes

Avacuum agent livesinatwo-squareworld(Left square, Right square), each square beingClean (C) or Dirty (D). Actions:Left,Right and Suck (Left in the left square and Right in the right square leave the agent where it is; Suckcleans the current square). Physical states are written (Agent, Left square, Right square):

State	Description	State	Description
S1	(L, D, D)	S5	(L, C, D)
S2	(R, D, D)	S6	(R, C, D)
S3	(L, D, C)	S7	(L, C, C)
S4	(R, D, C)	S8	(R, C, C)

(a) Initiallytheagent has no sensors and does not know its location or the dirt status. Formulate the problemover belief states: initial belief state, actions, goal test and path cost. **[3 Marks]**

(b) Applytheaction sequence [Right, Suck, Left, Suck] and write the belief state after each action. **[6 Marks]**

Step	Action	Belief State after Action
0	(start)	{S1, S2, S3, S4, S5, S6, S7, S8}
1	Right	
2	Suck	
3	Left	
4	Suck	

(c) Now the agent has a local sensor that reports [location, status of the current square]. Starting again from the initial belief state, the first percept is [Left, Dirty]; the agent then performs Suck and perceives [Left, Clean]; it then performs Right and perceives [Right, Dirty]. Fill in the belief state after each action (prediction) and after each percept (update). **[8 Marks]**

Step	Action	After Action (Predicted)	Percept	After Percept (Updated)
0	(start)	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	
1	Suck		[Left, Clean]	
2	Right		[Right, Dirty]	

(d) How many different belief states are possible when there are 8 physical states? Why does this make the problem harder than searching over physical states? **[3 Marks]**

(e) Interpret: explain why the percepts in part (c) shrink the belief state, how this differs from the sensorless case in part (b), and why the sensorless plan can still be called rational. **[5 Marks]**

Answers – Question 4

(a) Belief-state formulation

Initial beliefstate:{S1,S2,S3,S4,S5,S6,S7,S8}. Actions: Left, Right, Suck. Goal test: every possible physical state in the belief state is clean, i.e. {S7,S8}. Path cost: 1 per action.

(b) Sensorless action sequence [Right, Suck, Left, Suck]

0: {S1,S2,S3,S4,S5,S6,S7,S8}
1 Right: {S2,S4,S6,S8}
2 Suck: {S4,S8}
3 Left: {S3,S7}
4 Suck: {S5,S7}

(c) Prediction and percept update

- 0: Start belief{S1,S2,S3,S4,S5,S6,S7,S8}; percept [Left, Dirty] → updated {S1,S3}.
- 1 Suck: predicted {S5,S7}; percept [Left, Clean] → updated {S5,S7}.
- 2 Right: predicted {S6,S8}; percept [Right, Dirty] → updated {S6}.

(d) Number of possible belief states

2^8= 256 possiblebelief states. The problem is harder because the search space is the power set of the physical states rather than just the 8 physical states.

(e) Interpretation

Each percept removes physical states inconsistent with the observed location and dirt status, so the belief state shrinks. In the sensorless case, no percepts are available, so uncertainty remains and actions must work for every possible physical state. The sensorless plan is still rational because it guarantees that all possible states reach a clean configuration, maximizing the expected/guaranteed performance given the available information.

Rubrics forCalculation

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Search Problem	**20**	Clearly understands the search problem, initial p state, goal state, and search strategy.	Understands the problem with minor misunderstandings.	Demonstrates partial understanding of the search problem.	Showslittleor no understanding of the search problem.
State Expansion and Dry Run Execution	**30**	Correctly traces every P iteration, expands nodes accurately, and maintains the Queue/Stack/Priority Queue without errors.	erforms most iterations correctly with minor mistakes in state expansion or data structure updates.	Performs only some iterations correctly; several errors in tracing or node expansion.	Unable to correctly perform the dry run or expand states.
Application of Search Algorithm	**20**	Applies the selected algorithm (BFS/DFS/UCS) correctly, following all algorithmic rules.	Applies the algorithm correctly with a few procedural errors.	Applies only basic algorithm steps with noticeable mistakes.	Fails to apply the correct search algorithm.
Accuracy of Solution Path	**20**	Identifies the correct traversal order/solution pathandgoalstatewithmost proper justification.	Solu tionpathis ly correct with m ors.	Partiallycorrect solutionpat h; goal maynotbereached correctly.	Incorrect or incomplete solution path.
Accuracy of Solution Path	**10**	Queue/Stack/Priority Queue updates, state diagrams, and explanations are neat, organized, and logically presented.	Presentation is clear with minor and organizational issues.	Limited explanation organization; reasoning is partially clear.	Poor presentation with unclear or missing reasoning.

Q. No. / Criteria	Understa nding of Search Problem	State Expansion and Dry Run Execution	Application of Search Algorithm	Accuracy of Solution Path	Accuracy of Solution Path	Sub. Total (20)	Total %
1							
2							
3							
4							
Total							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____